

Red Hat, CentOS and Fedora, or *dpkg* in Debian and Ubuntu), Portage downloads and compiles the source code according to your preferred *USE* flags, and the instructions are in a file called *ebuild*. *USE* flags are almost like specifying *--prefix* when running *./configure*, but are defined globally, and make it easy to take care of dependencies. There is a good example in a section from the Gentoo handbook:

"Enter the *USE* flags. Such a flag is a keyword that embodies support and dependency-information for a certain concept. If you define a certain *USE* flag, Portage will know that you want support for the chosen keyword. Of course, this also alters the dependency information for a package.

"Let us take a look at a specific example: the *kde* keyword. If you do not have this keyword in your *USE* variable, all packages that have optional KDE support will be compiled without KDE support. All packages that have an optional KDE dependency will be installed without installing the KDE libraries (as dependencies). If you have defined the *kde* keyword, then those packages will be compiled with KDE support, and the KDE libraries will be installed as dependencies.

By correctly defining the keywords, you will receive a system tailored specifically to your needs."

So how does Gentoo Linux work? In short, you build the system more or less from scratch. Installation itself is very unique; the default Gentoo installation only provides a system with a minimal set of software, and believe it or not, there is no standard installer! Getting started with Gentoo Linux requires a bit of practice, but once you get used to it, I am more than confident that you will never want to go back to binary!

Installation methods

As I mentioned earlier, Gentoo Linux does not have its own installer (they used to have one [0], but it is now deprecated), and it heavily depends on manual installation via the command line. For those who are used to standard binary distributions with graphical installers, Gentoo installation is probably the most confusing and time-consuming.

First, we need to choose an installation medium. You may use a minimal installation LiveCD [1] to boot a minimal Linux environment, choose any other GNU/Linux LiveCD such as Knoppix, or an existing GNU/Linux system on your computer. Just note that the installation must be on a different disk/partition than what is used by an existing installation.

In Gentoo Linux, there are two main components that make Gentoo what it is: stages and portage. A stage is a tar-ball archive that contains the minimal Gentoo Linux environment. Basically, there are stages from 1 to 3 (unless you intentionally build Stage 4). To use Stage 1 or 2, you will need to bootstrap the system, which is explained in the FAQ [2] in the Gentoo handbook. Gentoo no longer recommends the use of Stage 1 through 2, unless you

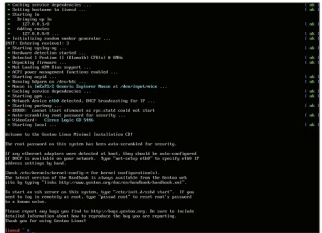


Figure 1: Booting from Gentoo minimal installation CD

are the release engineer and doing this for development purposes. For this article, we will use Stage 3, and install from the LiveCD, which is the standard Gentoo Linux installation procedure. Before going any further, it is not a bad idea to skim through the well-constructed Gentoo Linux handbook [3] to familiarise yourself with the stages and the portage.

Boot install CD

1. Download the ISO from <http://mirrors.kernel.org/gentoo/releases/x86/autobuilds/current-iso/> (filename: *install-x86-minimal-SDATE.iso*) and burn it as a bootable CD. If your computer does not have a CD drive, you can also follow the instructions at <http://www.gentoo.org/doc/en/liveusb.xml> to create a live USB on a spare USB stick.
2. Insert the CD (or USB stick) and reboot the machine. While still at the BIOS screen, press the special key (mostly F2 or Del) to go to *Set-up*, and configure the BIOS to boot from the CD (or USB stick) first. (Some BIOSs have short-cuts to the *Boot Device Priority* screen on pressing F12 or Esc.) The BIOS will soon read from the disk; once you see the SYSLINUX boot-loader, press RET and continue.
3. Once booted, you will be greeted with a nice frame-buffer console to work on. Most of the time, the LiveCD automatically detects and connects to the network. If not, use the console to configure the network like you usually do.

The install CD includes the OpenSSH daemon. If you prefer installing from a remote machine, or want to give others access to your installation environment (so that they can help you), set the root password first, and start the SSH daemon as follows:

```
[livecd ~]# passwd
Enter new UNIX password: <type password>
```